SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 – 2025

Follow For More Updates

Class : 11

COMMON QUARTERLY EXAMINATION - 2024 - 25

Time Allowed : 3.00 Hours] **CHEMISTRY** [Max. Marks : 70

PART – I

I. **Answer the following:** 15x1=15

1. The equivalent mass of a trivalent metal element is 9 g eq^{-1}, the molar mass of its anhydrous oxide is
 a) 102 g b) 27 g c) 270 g d) 78 g
2. Basicity of Sulphuric acid is
 a) 3 b) 1 c) 2 d) None
3. Splitting of spectral lines in an electric field is
 a) Zeeman effect b) Shielding effect c) Compton effect d) Stark effect
4. In a given shell the order of screening effect is
 a) s > p > d > f b) s > p > f > d c) f > d > p > s d) f > p > s > d
5. Atomic number of element having temporary name unniltrium is
 a) 101 b) 102 c) 103 d) 104
6. The cause of permanent hardness of water is due to
 a) $Ca(HCO_3)_2$ b) $Mg(HCO_3)_2$ c) $CaCl_2$ d) $MgCO_3$
7. The value of universal gas constant depends upon
 a) Temperature of the gas b) Volume of the gas
 c) Number of moles of the gas d) Units of pressure and volume
8. The pressure - density relationship $P_1/d_1 = P_2/d_2$ derived from the law
 a) Charles law b) Gay-Lussac's law c) Avogadro's law d) Boyle's law
9. If one mole of ammonia and one mole of hydrogen chloride are mixed in a closed, container to form ammonium chloride gas, then
 a) $\Delta H > \Delta U$ b) $\Delta H - \Delta U = 0$ c) $\Delta H + \Delta U = 0$ d) $\Delta H < \Delta U$
10. If the entropy change of a liquid is 16 J mol^{-1} K^{-1}, and the boiling point is 27°C, then the molar heat of Vapurisation is
 a) 4.8 KJmol^{-1} b) 4.6 KJ mol^{-1} c) 4.3 KJ mol^{-1} d) 4.2 KJ mol^{-1}
11. Which one of the following is incorrect statement?
 a) For a system at equilibrium, Q is always less than the equilibrium constant
 b) Equilibrium can be attained from either side of the reaction.
 c) Presence of catalyst affects both the forward reaction and reverse reaction to the same extent.
 d) Equilibrium constant varied with temperature.
12. Ortho and Para-nitro phenol can be separated by
 a) Azeotropic distillation b) Destructive distillation
 c) Steam distillation d) Cannot be separated
13. Which of the following compound exhibits optical isomerism?
 a) Dimethyl ether b) Acetic acid c) Lactic acid d) 2-butene
14. **Assertion** : Tertiary carbocations are generally formed more easily than primary carbocation.
 Reason : Hyper conjugation as well as inductive effect due to additional alkyl group stabilize tertiary carbonium ions.
 a) Both assertion and reason are true and reason is the correct explanation of assertion.
 b) Both assertion and reason are true but reason is not the correct explanation of assertion.
 c) Assertion is true and reason are false. d) Both assertion and reason are false.
15. What is the hybridisation state of benzyl carbonium ion?
 a) sp^2 b) spd^2 c) sp^3 d) sp^2d

PART - II

II. **Answer any six questions. Question No. 24 is compulsory.** 6x2=12

16. Define Equivalent mass?
17. State Pauli exclusion principle?

18. Why halogens acts as oxidising agents?
19. Give the uses of hydrogen peroxide?
20. What is compressibility factor?
21. What is lattice energy?
22. What is the relation between K_P and K_C. Give one example for which K_P is equal to K_C?
23. What is heterolytic fission?
24. Briefly explain geometrical isomerism in alkene by considering 2 - butene as an example.

PART - III

III Answer any six questions. Question No. 33 is compulsory. **6x3=18**

25. How many moles of hydrogen is required to produce 10 moles of ammonia?
26. Explain Davison and Germer experiment?
27. Ionisation potential of "N" is greater than that of "O". Give appropriate reasons?
28. What is Ortho and Para hydrogen?
29. Derive the relation between ΔH and ΔU for an ideal gas. Explain each term involved in the equation.
30. What are the various steps involved in Crystallisation?
31. Complete the following reactions.
    a) $CH_3Br + KOH \longrightarrow$
    b) $CH_3OCH_3 + HI \longrightarrow$
    c) $C_6H_6 \xrightarrow{Pt/H_2}$
32. State reaction Quotient?
33. An unknown gas diffuses at a rate of 0.5 time that of nitrogen at the same temperature and pressure. Calculate the molar mass of the unknown gas.

PART - IV

IV Answer all the questions. **5x5=25**

34. a) Explain the following redox reactions with an example.
    i) Decomposition reaction. (2½) ii) Metal displacement reaction. (2½)
    (OR)
    b) i) Give the electronic configuration of chromium and copper? (2)
    ii) How many radial nodes for 2s, 4p, 5d and 4f orbitals exhibit? How many angular nodes? (3)
35. a) i) Define Ionisation Potential? (2)
    ii) Explain the periodic trend of Ionisation potential? (3)
    (OR)
    b) What is Hydrogen bonding? Explain the types of hydrogen bonding with example.
36. a) i) State Graham's law of diffusion? (2)
    ii) What are Ideal Gases? In what way. real gases differ from ideal Gases? (3)
    (OR)
    b) State the various statements of second law of thermodynamics. (5)
37. a) Derive the relation between K_P and K_C. (5)
    (OR)
    b) i) State law of mass action? (2)
    ii) Give the IUPAC names of the following compounds. (3)
    a) $CH_3-\underset{CH_3}{\underset{|}{CH}}-\underset{Br}{\underset{|}{CH}}-CH_3$ b) CH_3-O-CH_3 c) CH_3COOH
38. a) Define optical Isomerism? What are the conditions for optical activity? (5)
    (OR)
    b) List out the postulates and dimitations of Bohr's Atom model. (5)